# Pre and Post-COVID San Francisco Bay Area Job Trends
## Final Report

Berkeley City & Regional Planning

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## Problem and Motivation

Studies have indicated two major issues arising in the post-pandemic landscape in the San Francisco (SF) Bay Area: (1) the workforce did not fully recover from the impact of the COVID-19 pandemic, and (2) the recovery has been extremely uneven across geographies and has deepened existing inequities (Levy, 2022). Though most people living and/or working in the Bay Area felt the consequences of the COVID-19 pandemic in the form of office closures, social isolation, and financial hardship, understanding where and how these difficulties are most pronounced is relevant, especially for policymakers and planning authorities. With Forbes Magazine naming San Francisco and San Jose as the #1 and #7 cities for small businesses in the United States (Arora, 2023), we also seek to understand which sized businesses have fared better than others over the past few years.

In 2025, the Metropolitan Transportation Commission (MTC) published the Vital Signs Workforce Report, which highlighted uneven sectoral changes across the metropolitan region. This study aims to add an additional layer of analysis by looking at changes in the current recovery process across different business/firm sizes and socioeconomic groups as well as to provide an updated report on the overall economic recovery across all the nine SF Bay Area counties. It does so by comparing employment-related data from before and after the pandemic, and provides more specific insights on recovery through three sub-questions, each targeting a distinct dimension of employment recovery post-pandemic by observing changes in the job recovery rate and jobs per capita indices:

1. **Post-pandemic Recovery:** What is the current state of employment recovery across the nine San Francisco Bay Area counties?
2. **Race:** How do jobs per capita differ across SF Bay Area counties by race in the post-pandemic period?
3. **Firm Size:** How did employment recovery (i.e. recovery index) rates differ across businesses of different sizes in the Bay Area?

Key terms used to define and frame the study:

1. **Geography:** Keeping project scope and feasibility in mind, the study is restricted to analyzing differences between the nine counties of the SF Bay Area (Alameda, Contra Costa, Marin, Napa, San Francisco, San Mateo, Santa Clara, Solano and Sonoma).
2. **Timeline:** 2018 - pre-pandemic baseline. 2024 - post-pandemic recovery timeline.
3. **Employment:** The number of jobs held on both the first and last day of the quarter with the same employer, i.e. stable employment, as reported by the Longitudinal Employer Household Dynamics (LEHD) Quarterly Workforce Indicators ("EmpS"). In this report the EmpS were chosen from the first quarter of each year starting from 2018.
4. **Workforce and Population:** The demographic distribution of workers by age and race/ethnicity as defined in the LEHD and American Community Survey (ACS) data.
5. **Firm Size:** A classification of employers by total number of employees as defined by the LEHD in categories of 0–19, 20–49, 50–249, 250–499, and 500+ employees.

6. **Recovery Index:** The ratio of the total number of employed workers (absolute employment) per county in a given year with respect to the baseline number of employed workers in 2018 in that specific county.
7. **Jobs per Capita Index:** The ratio of the number of employed workers per county with respect to the population of each county, calculated for each race and year from 2018 to 2024.

The primary unit of analysis is the county-level employment count (EmpS) within each sub-question category (there is no census tract level population data in LEHD). We used the absolute employment numbers to calculate two indices: a recovery index and a jobs per capita index (defined above), which are central to the visualizations. Given that the LEHD and ACS datasets are both at the county level, we acknowledge that the results of our analysis are subject to the Modifiable Areal Unit Problem (MAUP) bias, where patterns and relationships may vary at different levels of aggregation. County-level data may mask important distinctions between census tracts or neighborhoods, and may make certain sweeping generalizations about localized employment dynamics for the entire county. Despite the potential bias, this level of aggregation and granularity is chosen for the study since it allows us to calculate recovery and jobs per capita indices accurately using both datasets. It enables us to visualize differences between counties that may be relevant for stakeholders that work at the regional and county scale, such as regional planning agencies and county governments. MAUP, therefore, is not to be viewed as an inherent limitation of the study, but as an important point to be considered while making any inferences from the analysis.

As highlighted, we believe that the key stakeholders for whom this study can have an impact are regional planning agencies (like MTC), county governments, and community organizations. The findings may help in prioritizing workforce development programs, business support initiatives, and economic resilience planning toward the firm sizes and racial/ethnic groups with the lowest jobs per capita or recovery rates. By identifying where and for whom recovery has stalled, and which groups remain systematically underserved, this study aims to inform more equitable, data-driven economic support across the Bay Area.

## Data Description

Our research relies primarily on two CSV files: 1) the Longitudinal Employer-Household Dynamics (LEHD) Quarterly Workforce Indicators for employment numbers, time periods, location, and firm sizes, and 2) ACS 5-year estimates (2018-2024) for county level population. As Figure 1 indicates, in terms of coverage, the datasets span the 9 counties of the Bay Area at quarterly temporal granularity, with our research focusing on the time period defined earlier. Spatially, the LEHD data is aggregated at the county level, which is the same level as the ACS population data so that the jobs per capita index can be calculated accurately. The data is tabular in format (CSV), and has both numerical data such as discrete employment counts (EmpS) and categorical data such as race, industry code, and firm size tier. By merging it with the US Census ACS and TIGER/line shape file for the counties, we later convert this to a GeoDataFrame for mapping.

**Figure 1:** Data Description Table

| | **LEHD Quarterly Workforce Indicators** | **US Census ACS 5-Year Estimates** | **TIGER/Line Shape File** |
|---|---|---|---|
| **Coverage** | Nine counties of the Bay Area from 2018 to 2024 | Nine counties of the Bay Area from 2018 to 2024 | Nine counties of the Bay Area |
| **Granularity** | Quarterly temporal granularity spatially aggregated at the county level | Annual 5-year estimates spatially aggregated at the county level | County level spatial boundaries (no temporal granularity) |
| **Format** | Tabular (.csv) | Tabular (.csv) | Shape file (.shp) |
| **Data Type** | Numerical: EmpS<br>Categorical: race, firm size | Numerical: population for different races and ethnicities | Geospatial vectors: county boundary polygons/multipolygons |

*This table describes the LEHD, ACS, and TIGER/line shape file data in a comprehensive manner through four parameters: coverage, granularity, format, and data type.*

## Data Cleaning

A non-exhaustive summary of data cleaning steps and sanity checks conducted on the two datasets:

**LEHD Data Cleaning:**

1. Import: Brought in LEHD data as a .csv
2. Copy: Made a copy of the raw data for further cleaning.
3. Column names: Renamed all column names using snake case convention.
4. Maintaining necessary columns only: Dropped unnecessary columns to reduce clutter.
5. Data type check: Checked data types of each column to identify if they need to be corrected.
6. Fixed certain data types: changed employment numbers to numeric data, geography to string, and year to the date-time format. Most columns need to be retained as categorical.
7. Maintaining necessary columns only: Simplified the names of the new/fixed columns and dropped the old columns.
8. Duplicate cleaning: Dropped duplicates (if present) and cross checked to ensure every row is unique by checking for the sum of duplicate values.
9. Sanity check: Using df.head(), df.info(), and df.shape() several times to check the dataframe.
10. Save as processed data: Saved the cleaned dataframe into a csv.

**ACS Data Cleaning:**

1. Import: Brought in ACS data from Table B03002 for years 2018-2024.
2. Copy: Made a copy of the raw data for further cleaning.
3. Column names: Renamed all column names using snake case convention.
4. Maintaining necessary columns only: Dropped unnecessary columns (like Margins of Error) to reduce clutter, kept only the populations required for analysis.
5. Data type check: Checked data types of each column to identify if they need to be corrected.

6. Fixed certain data types: changed year to the date-time format for consistency. Most columns need to be retained as categorical.
7. Maintaining necessary columns only: Simplified the names of the new/fixed columns and dropped the old columns.
8. Duplicate cleaning: Dropped duplicates (if present) and cross checked to ensure every row is unique by checking for the sum of duplicate values.
9. Sanity check: Using df.head(), df.info(), and df.shape() several times to check the dataframe.
10. Save as processed data: Saved the cleaned dataframe into a csv.

**Missing Data Analysis:**
**Type 1 & 7:** ACS population estimates are not reflective of a specific year, and are using 5-year estimates which smoothens the overall population changes.
**Type 2:** With regards to data that we might not completely know why, the only relevant missing values were those for the absolute numbers of employed people. By checking the number of these missing values we found that of the whole dataset only approx. 6% were missing. This is probably due to the fact that some data was suppressed. These explicit missing values were dropped only when calculating the Recovery Index and Jobs per Capita. This might introduce bias to our estimates and we aim to investigate further the source of these.
**Type 3:** This analysis is focusing only on the 9 Bay Area counties. Limiting the scope is a necessary step for project feasibility and readability but might bias our interpretation of overall job trends.
**Type 4 & 11:** The LEHD dataset takes into consideration only formal jobs which we know have a different distribution by race. For example, this dataset does not provide us any information on the informal job market. The ACS dataset instead is based on survey responses that bring further bias. We acknowledge that underserved communities might be underrepresented in these datasets. In addition, smaller populations such as "Native" do not provide enough insight given their small sample size.
**Type 5:** Our proxies for job recovery (Recovery Index) and availability of jobs (Jobs per Capita) do not reflect overall job wellbeing; we haven't accounted for changes in job satisfaction or earnings over the years.
**Type 6:** Commuting flows, informal employment were not included in the analysis and could answer some of the assumptions raised in this paper.
**Type 8:** Statistical distinctions between race and ethnicity led to separate (but consistent) calculations when accounting for the Hispanic population.
**Type 9:** County-level aggregation simplifies our conclusions (see MAUP discussion).
**Type 10:** Survey-based ACS estimates introduce uncertainty in the precision of recovery and jobs-per-capita measurements.
**Type 11:** Not directly applicable.
**Type 13:** The Native American population was omitted from the analysis given their small sample size.
**Type 14:** Not directly applicable.
**Type 15:** The findings of this research should not be generalized to all regions or populations.

## Methodology

As discussed earlier, this analysis consists of two parts: a recovery index and a jobs per capita index. Moreover, the analysis has been further divided to look at these indices across races/ethnicities as well as firm sizes. To start with, we computed recovery indices for all counties over the entire timeframe of 2018-2024, allowing us to visualize change over time. To pull out county-level distinctions, we then generated a choropleth map specifically for 2024 to understand recovery patterns post-pandemic. In continuation of overall

regional recovery, we looked at differences across firm sizes. Detailed analysis focused on jobs per capita distinctions by race and ethnicity, particularly four groups: White, Black, Asian, and Hispanic populations. As discussed under the missing data section, Native, Alaska, and Pacific Islander populations have been excluded due to very small sample sizes. A more detailed description of the data analysis:

**Calculating employment recovery and jobs per capita indices by county and firm size:**

1. Filtered for all industries and all owner codes.
2. Computed the Recovery Index (using 2019 as baseline) on the *df_totals* dataframe.
3. Standardized geographic identifiers (county) to enable consistent merging with LEHD data.
4. Merged LEHD and ACS data using a unique identifier: "county".
5. Created interactive line charts of Jobs per Capita using Plotly Express based on both county and firm size.
6. Added TIGER/line county shape files using geopandas and constructed GeoDataFrames.
7. Created a choropleth map for 2024 for Recovery Index (all races) by county.

**Calculating jobs per capita differences across Bay Area counties by race and ethnicity:**

1. Filtered for all industries, all firm sizes and all owner codes.
2. Merged LEHD and ACS data on a new dataframe *df_race* and grouped by county and race.
3. Computed the Jobs per Capita index.
4. Created line charts using race-specific recovery indexes by county (sampled), made interactive using Plotly Express.
5. Added TIGER/line county shape files using geopandas and constructed GeoDataFrames.
6. Created a choropleth map for 2024 for Jobs per Capita by county for each race group.
7. Repeated all steps for ethnicity map (Hispanic) as it falls under a separate category as per the ACS.

## Results

**Figure 2:** LEHD Summary Statistics

| | Count | Mean | Standard Deviation | Minimum | 25% | 50% | 75% | Maximum |
|---|---|---|---|---|---|---|---|---|
| **Employment Data (EmpS)** | 139496.0 | 2523.29 | 17809.24 | 0.0 | 14.0 | 90.0 | 664.0 | 963429.0 |

*Summary statistics of the LEHD data provides a descriptive overview of the primary unit of analysis, i.e. the stable employment data (EmpS). It is important to note that this summary statistics takes into account all sectors, races, and counties. The relevant piece of information provided in this table is that the data cleaning process was carried out correctly (minimum value = 0).*

**Part 1: Recovery Index Analysis**

Preliminary analysis indicates that all Bay Area counties experienced a significant drop in total employed people during the pandemic years; they have all largely recovered since then, but the recovery has been uneven. In fact, the chart depicting absolute employment rate by county over time shows that, apart from San Francisco, all counties saw an increase in post-pandemic employment compared to the 2019 baseline. Figure 3 is pre-selecting the San Francisco and Alameda counties and shows how San Francisco is

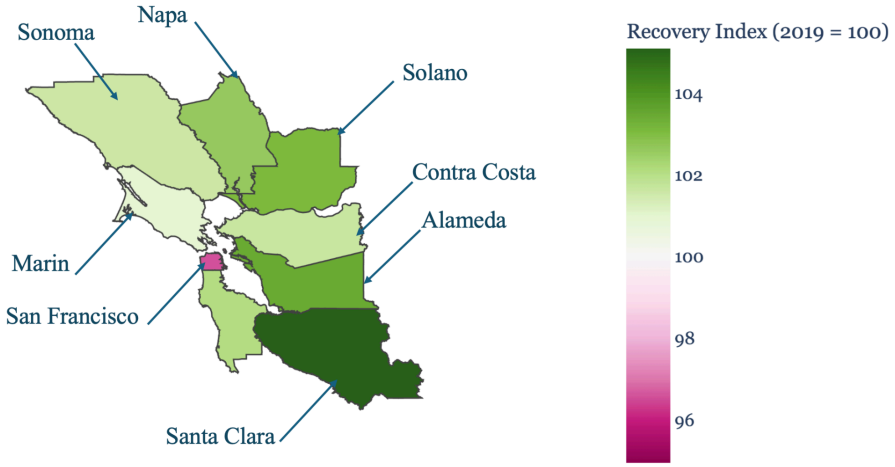
experiencing a slower recovery than relatively more suburban counties such as Alameda. This is supported by the literature, since core city centers tend to have a higher share of service jobs, more exposed to the impacts of the COVID-19 pandemic and the shift to remote working (Pardue, 2021).

**Figure 3**: Recovery Index by County (Base Year = 2019, County = Alameda, San Francisco)

*The interactive graph, available on the ArcGIS story map, allows you to choose and compare the recovery index trends for all 9 San Francisco Bay Area counties. Normalization: Recovery index made comparable by calculating values relative to a baseline year of 2019. Aggregation: County-level across all races/ethnicities, firm sizes, and industry types across all years 2018-2024. Source: ACS and LEHD data (2026).*

This is further supported by the choropleth map across all races in each county, specifically for the year 2024 (Figure 4). A ratio or index of 100 is considered as the baseline (2019), with values above 100 indicating an increase in total jobs per county (compared to the baseline) and values below 100 indicating a decrease. Figure 4 shows that San Francisco County is the only county that has not yet recovered from pre-pandemic employment levels, as of 2024. The county that experienced the highest recovery is Santa Clara County (Recovery Index above 104).

**Figure 4:** Bay Area County Recovery Index as of 2024 by County

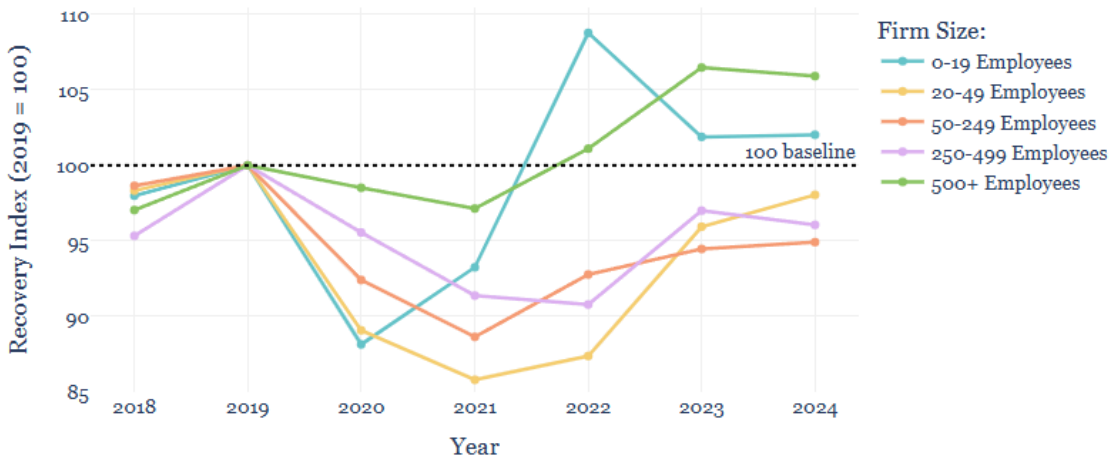



*This interactive illustration includes the 9 SF Bay Area counties, wherein any county with a recovery index above the baseline is color-coded green, while counties below the 100 baseline are shown in purple to improve readability. The*

*magnitude of recovery is represented using a continuously graduated divergent colorblind-safe color scheme. Normalization: Recovery index made comparable by calculating values relative to a baseline year of 2019. Aggregation: County-level across all races/ethnicities, firm sizes, and industry types for 2024. Source: ACS and LEHD data (2026).*

With regards to firm size, LEHD data shows that small (0-19 Employees) and large firms (500+ employees) have recovered from pre-pandemic levels (Figure 5). Differently, medium-sized firms (20-499 Employees) do not seem to have yet recovered. Very large firms (500+ employees) also appear to be the most financially healthy enterprises in the Bay Area, confirming that greater access to capital and international markets contributes to stronger resilience. This is partly supported by the literature. Chandler et al. (2021), by analyzing post-pandemic recovery trends in 100 US metropolitan cities, showed that medium-sized firms were the hardest hit by the pandemic in the medium-term due to the limited access to capital. Unlike the report's findings, small firms in the SF Bay Area appear to have largely recovered. Further research could examine whether these differences are consistently observed between urban and suburban counties. Very small firms such as local shops might have recovered faster in suburban counties compared to service-oriented cities such as San Francisco.

**Figure 5:** Recovery Index by Firm-size across all SF Bay Area Counties



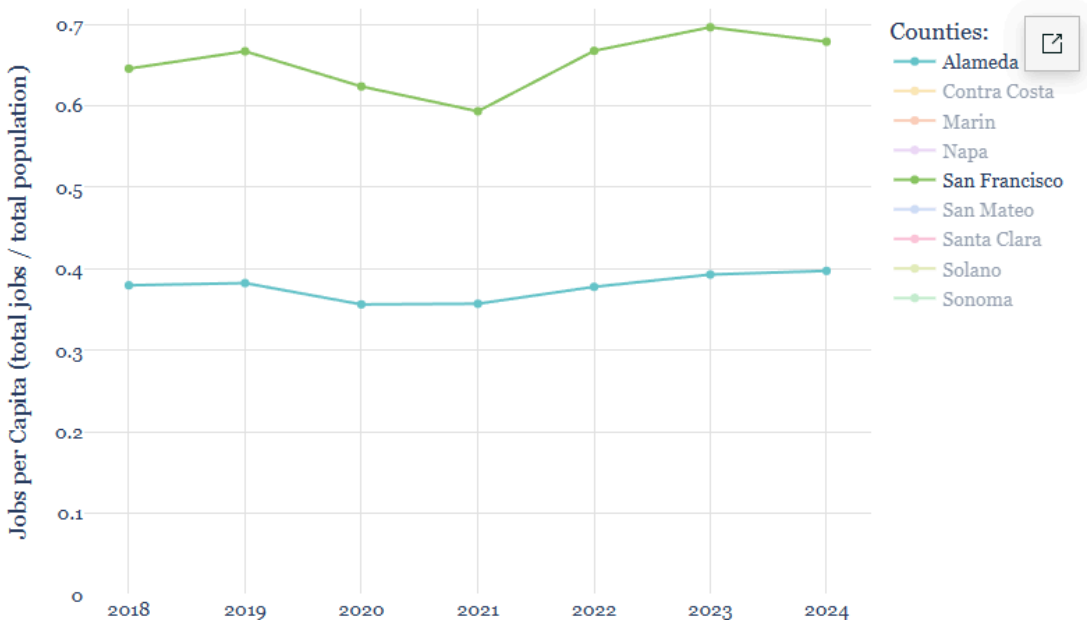

*This illustration, interactive in the ArcGIS version of the report, shows how the recovery index changed relative to the 2019 baseline depending on firm size. Firms with 0–19 employees are considered very small, those with 20–249 employees are classified as medium-sized, and firms with 250 or more employees are considered large. Given the limited number of firm-size categories available in the dataset, the data was not further aggregated into broader employee bins. Normalization: Recovery index made comparable by calculating values relative to a baseline year of 2019. Aggregation: Regional-level/Bay Area-level across all counties, races/ethnicities, and industry types, focusing on distinctions between firm sizes across all years 2018-2024. Source: ACS and LEHD data (2026).*

**Part 2: Jobs per Capita Index Analysis**

The jobs per capita index by county graph (Figure 6) tells us that although the recovery index was lower for San Francisco County compared to the 2019 baseline, it is doing considerably better than the other counties in terms of the ratio of total employment to total population. However, it is important to note that a ratio of, say, 1.0, does not mean that these jobs are directly serving the residents of that particular county itself - it merely indicates that there are as many employed workers/jobs as there are people in that county. Future

research could investigate how more suburban counties next to the Bay Area perform in terms of jobs per capita so as to better understand whether the assumption above is supported by further evidence.

**Figure 6**: Jobs per Capita by County in the SF Bay Area (Alameda & San Francisco)




*This illustration shows how more suburban counties such as Alameda have a considerably lower jobs per capita index (0.4) than more service-oriented counties such as San Francisco (0.7). Figure 6 can be pre-selected in the ArcGIS interactive version in order to compare jobs per capita for different counties. Normalization: Jobs per capita index made comparable by calculating employment numbers relative to the population estimates. Aggregation: County-level across all races/ethnicities, firm sizes, and industry types across all years 2018-2024. Source: ACS and LEHD data (2026).*

Next, we looked at the jobs per capita for four races/ethnicities across all the nine Bay Area counties. Figure 7 shows how predominantly suburban counties such as Solano and Contra Costa seem to have the lowest jobs per capita indices across all races, possibly implying that these counties have the fewest jobs directly situated within the area. There are also variations by county and race (Figure 7); San Francisco and Santa Clara County have the highest jobs per capita for white alone, possibly due to the presence of the tech industry. For black alone, Solano, Contra Costa, and Alameda stand out as having the lowest jobs per capita index. This is notable due to the fact that those three counties have the highest percentage of Black residents in the Bay Area (Bay Area Census, 2026). Differently, Marin and Napa show to have the highest jobs per capita for the Asian population. The Hispanic population map tells a drastically different story: only San Francisco County has a relatively high jobs per capita index of 0.8, while all other counties range between 0.4 to 0.5. This could be explained by the higher employment in low-wage/informal sectors for this population group (US Bureau of Labor Statistics, 2023). This has implications for commuting costs and burdens on specific races and ethnicities. Future research could investigate potential correlations between jobs per capita by county and race with commuting costs, public transit routes, and household earnings.

**Figure 7**: Jobs per Capita by County for Race/Ethnicity - White, Black, Asian, Hispanic

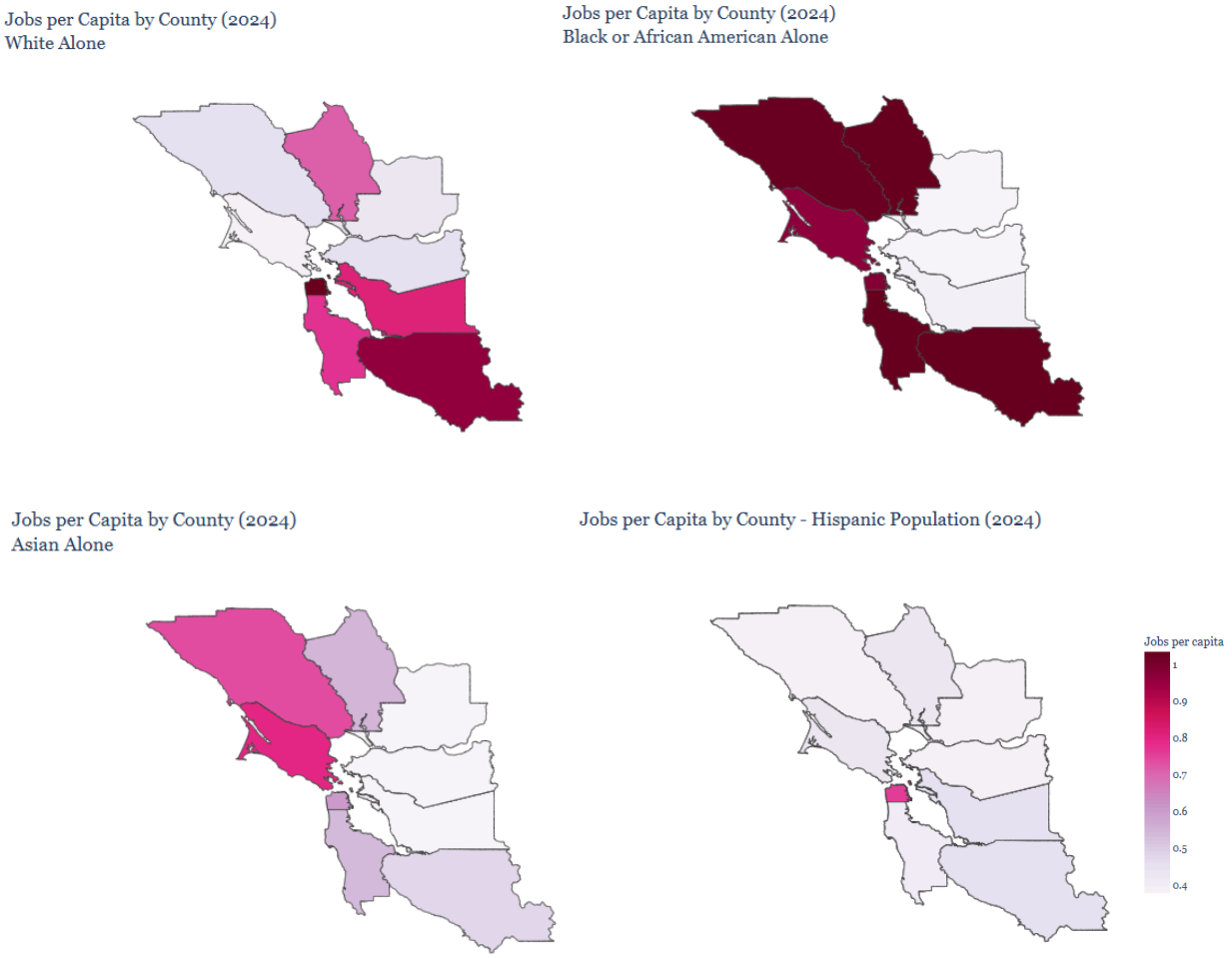



*Figure 7 shows that across all counties, white alone have the overall highest jobs per capita index as compared to Black, Asian, and Hispanic populations. The magnitude of jobs per capita is represented using a continuously graduated colorblind-safe color scheme, wherein darker colors indicate higher index values. Normalization: Jobs per capita index made comparable by calculating employment numbers relative to the population estimates. Aggregation: County–and-race-level across all firm sizes and industry types for 2024. Source: ACS and LEHD data (2026).*

In order to have a more detailed picture on a sample of two countries, we looked at the jobs per capita (across all races) for Alameda and San Francisco. Figure 8 indicates that in San Francisco County, white alone have the highest jobs per capita index while other races are below 1.0. In Alameda County, on the other hand, the highest index is 0.8 (white alone), and other races are closer to half the number of jobs per capita as white alone (0.4). This suggests that all races, apart from the white population, possibly rely on jobs located outside of the county of origin. Further research is needed to show any causal connections between the place of residence and employment. The common thread for all counties appears to be that white alone consistently fares better than other races.

**Figure 8**: Jobs per Capita by Race (San Francisco County and Alameda County)

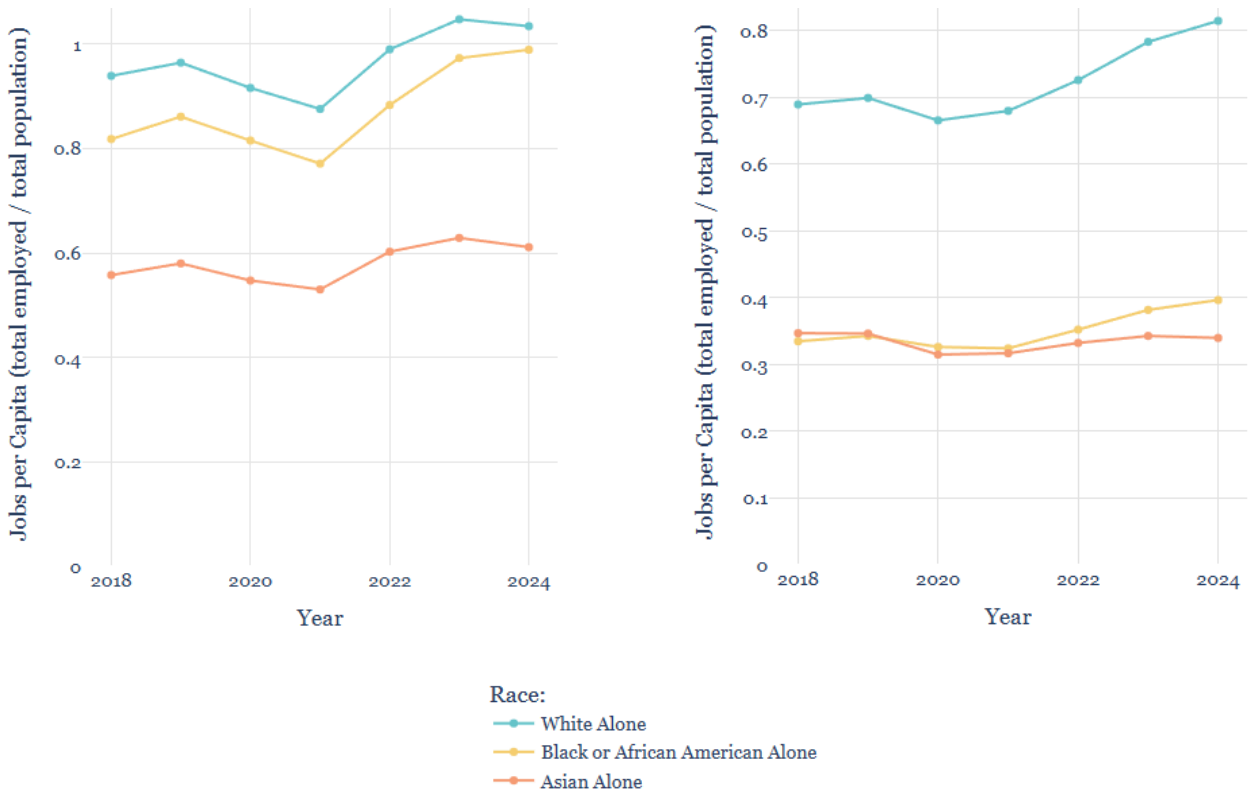



*Figure 8 shows that in San Francisco County, white alone have the highest jobs per capita index while other races are below 1.0. In Alameda County, the highest index is 0.8 (white alone), and other races are closer to half the number of jobs per capita as white alone (0.4). Normalization: Jobs per capita index made comparable by calculating employment numbers relative to the population estimates. Aggregation: County–and-race-level across all firm sizes and industry types across all years 2018-2024. Source: ACS and LEHD data (2026).*

## Limitations

As discussed in the initial sections of the report, the analysis is undoubtedly subject to the Modifiable Areal Unit Problem (MAUP), meaning that different levels of spatial aggregation result in different results and interpretations. As this is a county-level analysis with a temporal granularity of only stable employment as reported by the LEHD, it may not be the most comprehensive analysis of minute nuances as it generalizes results across the entire county. Additionally, the LEHD dataset only focuses on the formal job market, leaving out several informal employment opportunities unaccounted for in this analysis. Similarly, self-employed individuals, domestic labor, and other precarious forms of employment are only partially represented or excluded entirely from LEHD records.

## Conclusion

The economy of the San Francisco Bay Area has transformed considerably across the past half decade. Though we did not run any tests of statistical significance on the data in this report, it is reasonable to assume that the COVID-19 pandemic was responsible for this transformation. This report builds on the work carried out by the MTC by understanding (i) the current state of employment recovery across the nine San Francisco Bay Area counties and amongst different firm sizes, and (ii) how jobs per capita rates differ across the Bay and by race. We find that (i) San Francisco is the only county that did not fully recover from the COVID-19 pandemic (ii) medium sized firms did not fully recover from the pandemic (iii) San Francisco has the highest jobs per capita in the Bay confirming its role as central economic hub and (iv) jobs per capita is the highest amongst the white population across all the Bay Area counties. We believe that race/ethnicity- and business-specific commuting patterns are central in understanding the Bay Area's job market. This report shows that post-pandemic economic recovery has not occurred uniformly across demographic groups or places within the region, but further research is needed to suggest any causal mechanisms. We recognize that other factors, such as political turnover and the commercial real estate market are also at play. Using the same geographic and demographic parameters, future research should examine the impact these factors have had on the economy to build a more holistic picture of the San Francisco Bay Area before and after the COVID-19 pandemic, providing valuable insights for the policy and planning community to stimulate healthy growth/recovery.

**Link to interactive ArcGIS StoryMap:** https://arcg.is/HKzvq2

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